Advanced Financial Modeling, Module 10: Take-Home Private Equity Case Study and LBO (Cars.com)

In this module, you'll complete an open-ended, "take-home" exercise based on Cars.com, which requires you to find and research a company, propose an operational turnaround or improvement, and then build an LBO model to support it.

This case study requires significant outside research to find suitable add-on acquisition candidates and to support the key drivers.

There's also a 15-slide investment recommendation presentation at the end.

AFM-10-01: Take-Home LBO Case Study Overview and Company Selection (21:41)

In this introductory lesson, you'll learn what to expect in take-home LBO case studies and how you might screen sectors and companies and pick a good candidate for the case study based on a private equity firm's investment strategy.

Table of Contents:

- **1:35:** Open-Ended Case Studies
- **4:24:** Company Selection Criteria
- **8:14:** Narrowing the Sector
- **11:39:** Picking a Specific Company
- **18:15:** What to Do Without Capital IQ
- **20:11:** Recap and Summary

Quiz Questions:

8:13 When selecting a company for an open-ended LBO modeling case study, which of the following is NOT an important or useful criterion?

a) **The company should have minimal existing Debt in its capital structure.**
b) The company should trade at a relatively low EBITDA multiple.
c) The company should consistently generate Free Cash Flow.
d) The company's financial statements should be relatively simple and straightforward.

Explanation: The first answer is correct, i.e., this criterion is NOT important when selecting LBO candidates. The issue here is that the company's existing capital structure is always wiped out and replaced when a deal takes place, so its pre-deal Debt level does not matter (you could argue it matters a bit if it means that lenders already know the company, but that's a tiny

benefit next to everything else here). The other criteria are all much more important for selecting a good LBO candidate.

AFM-10-02: Cars.com – Data Gathering and Financial Statement Overview (14:39)

You'll learn how to gather the required data from the company's filings and outside research and how to input and modify the financial statements for use in the LBO model in this lesson.

Table of Contents:

- **1:52:** Scan of the Annual Report
- **5:42:** Driver Projection Methods
- **9:01:** Outside Information Gathering
- **11:18:** Financial Statement Entry
- **13:10:** Recap and Summary

Quiz Questions:

11:17 Why does Cars.com depend more on the car dealership market than the total volume of new and used vehicle sales?

a) Because the vehicle sales do not change much from year to year, but the car dealership count does.
b) Because the company sells advertising primarily to car dealerships.
c) Because it earns most of its revenue by charging car dealers to use its services and connecting them with consumers.
d) All of the above.

Explanation: The third answer is correct. Since Cars.com sells its services to dealers and does not correct car buyers and sellers directly, the growth of the car dealership market makes more of a direct impact on the company's business than the volume of vehicle sales in the U.S. The first answer is wrong because neither vehicle sales nor the car dealership count change much in mature, developed countries, and the second answer is wrong because the company's main business is selling subscriptions to its marketplace to dealerships, not advertising.

AFM-10-03: Cars.com – Transaction Assumptions (13:36)

In this lesson, you'll set up the transaction assumptions for Cars.com, including the purchase premium, Enterprise Value bridge and purchase multiples, minimum Cash, scenarios, and the first part of the Sources & Uses schedule.

NOTE TO LESSON: At 5:19, we state that the company does not disclose the Face Value or Market Value of its Debt. While this is true, it does disclose later in the filings the unamortized issuance fees, so we can estimate the Face Value of the Debt and use it in the bridge. We fix this point in the next lesson.

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- **0:32:** Transaction Assumptions
- **10:34:** Sources & Uses, Part 1
- **12:28:** Recap and Summary

Quiz Questions:

10:33 When might you make the Minimum Cash balance a constant number in an LBO model?

a) If it has stayed in a similar range historically over many years.
b) If the company's core business is not expected to change much in the projected period.
c) If the company's cash conversion cycle is fairly quick.
d) All of the above.

Explanation: Everything above is true. If the company's business is not changing much, and its Cash balance has also been in the same range over time, it's reasonable to assume a constant number for the minimum Cash balance in the projections. A faster cash conversion cycle also helps because it means the company collects and spends cash quickly and doesn't need to retain as much of a balance to pay for ongoing operating expenses.

AFM-10-04: Cars.com – How to Make Your Own Debt Assumptions (13:00)

You'll learn how to read the company's notes about Debt in its 10-K and use the existing terms to develop new terms for the post-LBO Debt in this lesson, including differences in interest rates, amortization, and maturities.

Table of Contents:

- **2:17:** Finding the Current Debt Terms
- **6:38:** Proposing New Terms
- **9:00:** Sources & Uses, Part 2
- **11:14:** Recap and Summary

Quiz Questions:

8:59 Which of the following is the BEST reason to assume a slightly higher interest rate on the post-LBO Debt, assuming the company has Debt before the LBO?

a) **If the company's Total Leverage will be higher post-LBO.**
b) If the company will use its FCF to repay Debt rather than accumulate Cash post-LBO.
c) If the holding period is expected to be slightly longer than the current Debt maturities.
d) All of the above are equally good reasons.

Explanation: The first answer is the best choice. All else being equal, higher leverage means higher credit risk, which means that lenders will demand higher interest rates to fund the company. Using FCF to repay Debt rather than letting it accrue to the Cash balance reduces credit risk, if anything, and a slightly longer holding period should not make a huge impact on the interest rates next to the increased leverage.

AFM-10-05: Cars.com – Revenue Projections (11:29)

In this lesson, you'll project revenue for Cars.com in the different scenarios, and you'll learn how to make reasonable assumptions for drivers such as the market growth rate and the company's market share.

Table of Contents:

- **1:13:** Market Growth and Market Share
- **4:14:** Revenue per Dealer, Traffic, and Advertising
- **7:16:** Revenue in the Different Cases
- **10:00:** Recap and Summary

Quiz Questions:

9:59 Why does Cars.com's standalone revenue not change much over 5 years in the Base case of the LBO model?

a) This is a standard assumption for the Base case; the business should neither grow nor decline by much.
b) Because the company's growing advertising revenue offsets its falling subscription revenue.
c) Because the company's advertising revenue per visitor falls, even as its total website visitors increase.

d) **Because its falling dealership market share is offset by price increases on its subscription product.**

Explanation: The last answer is the best choice. The company's market share falls by 0.3% to 0.4% per year, but since its prices also increase by 2-3% per year, marketplace subscription revenue stays about the same, which is the key reason why its total revenue stays about the same. Stagnant revenue is not a "standard" assumption in the Base case, as it depends heavily on the company and industry, and Cars.com advertising revenue does not grow at all, so the second choice is wrong. The third answer is a correct statement, but it's not the key reason why total revenue stays about the same in the Base case because marketplace subscription revenue is far higher than advertising revenue.

AFM-10-06: Cars.com – Expense and Cash Flow Projections (21:40)

You will analyze the historical expenses and cash flow line items for Cars.com in this lesson and set up projections for them in the holding period, and you'll see how the margins and cash flows of the company's business compare in different scenarios.

Table of Contents:

- **2:19:** Income Statement Expenses
- **12:59:** Cash Flow Line Items
- **19:27:** Recap and Summary

Quiz Questions:

12:58 In an LBO model, what's the easiest way to ensure that the company's EBIT and EBITDA margins do not change dramatically in different cases?

a) Build a detailed schedule that links expenses to the employee count in each category.
b) **Make most of the expenses simple percentages of revenue.**
c) Make the expenses percentages of revenue and ensure that some increase and some decrease to offset them, depending on the case.
d) Set up a "plug" that adjusts the company's margins if they rise too high or fall below a certain level.

Explanation: The second answer is the best choice because if most expenses are simple percentages of revenue, the margins should stay about the same even if the company's revenue changes significantly in different cases. It's possible to use the other approaches described here, but they are not "the easiest ways" of ensuring margins in about the same range.

AFM-10-07: Add-On Acquisitions, Part 1 – Data Gathering (14:30)

In this lesson, you'll learn how to form baseline assumptions for the add-on acquisitions by reviewing the financial profiles of acquisition candidates on Capital IQ to get a sense of the growth rates, margins, EBITDA levels, and multiples.

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- **3:19:** Acquisition Ideas
- **8:54:** Acquisition Assumptions
- **12:56:** Recap and Summary

Quiz Questions:

[No quiz questions.]

AFM-10-08: Add-On Acquisitions, Part 2 – Revenue, EBITDA, and Financial Statement Impact (11:57)

You'll complete the add-on acquisition setup in this lesson by projecting the revenue, EBITDA, and EBIT from these deals and including them on the financial statement projections for Cars.com.

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- **1:51:** Revenue and EBITDA
- **7:11:** Financial Statement Integration
- **10:33:** Recap and Summary

Quiz Questions:

10:32 Which of the following conditions allow(s) us to simplify the treatment for add-on acquisitions in an LBO model?

a) If the LBO model does not include Balance Sheet projections.
b) If we care about only the *cash effects* of the acquisitions.
c) If we assume that each add-on acquisition has an identical financial profile.
d) All of the above.

Explanation: Everything above is true. Without full 3-statement projections (i.e., no Balance Sheet in an LBO), acquisition modeling is much simpler because you don't need to create Goodwill and Other Intangible Assets and amortize the Intangibles. Ignoring non-cash items such as this Amortization also simplifies the statements and makes the tax treatment easier. Finally, if each acquisition has the same financial profile, the schedule is easier to setup, and less data gathering is required.

AFM-10-09: Debt Schedule, Part 1 – Cash Flow, Term Loans, Subordinated Notes, and Revolver (13:39)

In this lesson, you'll set up a Debt Schedule for Cars.com that supports additional Term Loan issuances to fund the add-on acquisitions, and you'll see how the amortization and cash flow sweep formulas differ.

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- **2:54:** Cash Flow Surplus or Shortfall
- **5:10:** Term Loans
- **9:36:** Subordinated Notes
- **10:18:** Revolver
- **11:12:** Recap and Summary

Quiz Questions:

9:35 If a company keeps issuing Term Loans to fund add-on acquisitions in an LBO, should the principal repayment or amortization formula in each year reflect the new issuances in that year?

a) Yes, because principal repayment always begins the moment the new Term Loans are issued.
b) It's not necessary if the cash flow sweep formula captures the optional principal repayments.
c) No, because the issuances typically occur at the end of each year.

Explanation: The second answer choice is correct. If there are additional Term Loan issuances each year, the fixed amortization will change over time, but recording this change in the current year (i.e., beginning-of-year or mid-year issuance assumed) vs. the next year (end-of-year issuance) makes a tiny difference in the model. The most important point is that if additional issuances occur, and the company's cash flow is sufficient to repay them, then we must allow for that in the model via the cash flow sweep formula. The first answer is not correct because if

the loans are issued on December 31, principal repayments may not begin until the next year, and the last answer is wrong because the issuances could, in theory, occur at any time in the year.

AFM-10-10: Debt Schedule, Part 2 – Interest Rates, Financial Statements, and Metrics and Ratios (19:52)

You'll complete the Debt Schedule in this lesson, link the interest and amortization line items on the financial statements, and calculate the key metrics and ratios, such as FCF conversion, FCF yield, and Return on Invested Capital (ROIC) to draw some quick conclusions about possible multiple expansion in the deal.

Table of Contents:

- **1:09:** Interest Rates
- **5:21:** Financial Statement Integration
- **10:51:** Key Metrics and Ratios
- **18:31:** Recap and Summary

Quiz Questions:

18:30 Which of the following is the strongest indication that multiple expansion in an LBO might be plausible?

a) The company's FCF yield increases substantially by the end of the holding period.
b) The company's FCF conversion improves significantly over time.
c) The company's ROIC improves significantly by the end.
d) All of the above are equally strong indicators.

Explanation: The third answer choice is the best one. All else being equal, EBITDA multiples trend with EBITDA growth and the company's Return on Invested Capital (ROIC), i.e., how efficiently it is using the capital from all investors.

AFM-10-11: Exit Multiples and Returns Calculations (14:17)

In this lesson, you'll review the company's trading history, comparable companies' multiples, and the improvement in its financial metrics to estimate a reasonable range of exit multiples in different scenarios, and you'll calculate the IRR and multiple to the sponsor based on that.

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- **2:58:** Ranges of Exit Multiples
- **9:45:** Returns Calculations
- **13:05:** Recap and Summary

Quiz Questions:

13:04 Which of the following represent(s) typical DIFFERENCES between IPO and M&A exits in LBO deals?

a) It might take the PE firm several years to sell its entire stake in an IPO exit.
b) In an IPO exit, the company is often valued at a slightly higher exit multiple.
c) In an IPO exit, there's more uncertainty because the average exit multiple might change over time, and the time frame is unknown in advance.
d) All of the above.

Explanation: Everything above is true. These all represent typical differences when exiting an LBO with an initial public offering (IPO) rather than a merger or acquisition. Since it takes longer to sell the stake, the exit multiple could fluctuate, and the IRR might end up being lower even if the multiple of invested capital or money-on-money multiple is the same. Companies tend to be valued at premiums when they become independent, publicly traded companies as well, so some people assume slightly higher exit multiples to reflect that.

AFM-10-12: Sensitivity Tables (17:03)

You'll create sensitivity tables to analyze the Cars.com leveraged buyout and the add-on acquisitions in the holding period here, and you'll use these tables to draw a few quick conclusions about the deal and its risk factors.

Table of Contents:

- **1:28:** Standard Tables
- **7:16:** Add-On Acquisition Tables
- **15:30:** Recap and Summary

Quiz Questions:

15:29 What is the most logical reason that explains why larger add-on acquisitions might reduce the risk of add-on acquisition multiples increasing too much in an LBO?

a) Larger acquisitions directly justify higher exit multiples.

b) Larger acquisitions tend to boost growth rates and the ROIC more.
c) Larger acquisitions tend to be easier to execute and integrate.
d) Larger acquisitions make up for the lower EBITDA contributions at higher multiples.

Explanation: The last answer choice is correct, i.e., it has the most logical explanation. Since the Exit Enterprise Value is based on the Exit EBITDA and the Exit Multiple, anything that increases the Exit EBITDA, such as larger acquisitions, will tend to increase the Exit TEV and the IRR in the deal. So, if the average price increases from 5x EBITDA to 8x EBITDA, we could offset the lower EBITDA contribution by spending more on the acquisitions. The first two answers do not make sense because the add-on acquisition size does not directly determine the exit multiples, and larger deals do not necessarily translate into higher growth rates or ROIC figures. The third answer is wrong because larger acquisitions tend to be *more* difficult to execute and integrate.

AFM-10-13: Cars.com – Investment Recommendation Presentation (17:28)

In this lesson, you'll learn how to outline and draft a 15-slide investment recommendation presentation for this leveraged buyout of Cars.com, and you'll learn what the presentation and model do well and the areas that could use improvement.

Table of Contents:

- **0:45:** Recommendation Outline
- **5:09:** The Market
- **9:09:** LBO Model Output
- **11:47:** Risk Factors and Summary
- **14:55:** Critique of This Recommendation

Quiz Questions:

14:54 Why might it be problematic to recommend that a company invest more to grow one segment of the business as a way to mitigate risk in another segment?

a) We may not have details on the time required or upfront investment.
b) Without additional research, we have no way of knowing if other segments have growth potential.
c) If the extra investment does not result in EBITDA and FCF fairly soon, it might not offset the decline in the other segment.
d) **All of the above.**

Explanation: Everything above represents potential problems with this idea for mitigating risk. When you propose something like this, you should have at least basic numbers and data to back up how well this strategy might work and how much it might cost in time and money to implement.